From EO research to a useful public demo

Five scientific bets, three shortlisted workflows, and the evidence needed before building.

Three bets, three different strengths

Aim: a scientific tool people can question and use.
Forecast + observe is the strongest WAM fit. Coastal claim validity has the better near-term evidence path. Model-upgrade continuity remains supporting infrastructure.

Best match to the long-term aim

Forecast + observe

`Gate: timestamps, targets, rights`

Best first physical instrument

Coastal claim validity

`Gate: close-time survey matches`

Clearest search interaction

Burn analogue search

`Gate: verified relevance + no-match`

These are recommendations, not demonstrated impact, robustness, savings or adoption.

WAM = world-action model · S1 = Sentinel-1 radar · S2 = Sentinel-2 optical. Here, actions change available information—not Earth.

Bet	Evidence already available	What must still be proved
Forecast + observe	Existing vegetation forecasts and consequential warning workflows	Task-aware selection beats newest/clearest-scene rules without leakage
Coastal claims	Published beach surveys and documented manual screening	More reliable physical claims than conservative error propagation
Burn search	A historical fire archive and strong spectral/metadata filters	Verified cross-site analogues and reliable rejection of absent cases

01 / Scientific bet

Forecast change. Choose useful evidence.

● Conditional · data gate unresolved

Which available radar or optical scene would improve a 20–30-day greenness forecast enough to justify processing it?



The possible contribution

Predict **task-specific observation value** under missing sensors and geographic/time shift—not just future pixels or generic uncertainty.

Data starting point

GreenEarthNet for optical forecasting; EarthNet2023 Africa is a candidate for multisensor replay. Native scene chronology, future targets and rights are unverified.

What success would have to show

Better proper forecast scores per processing cost than cheap scene-selection rules, with honest calibration on held-out sites/dates.

Why it might fail

No usable radar/optical chronology; obscured target truth; future-weather leakage; or newest/clearest scenes work just as well.

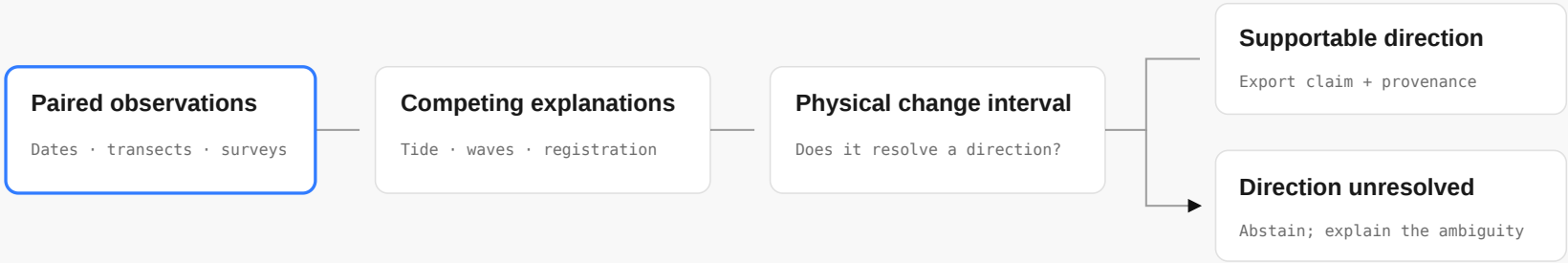
Space scope: curated historical replays first. This is not a live global forecast, satellite tasking or land-management intervention.

Already exists: Contextformer, probabilistic NDVI forecasting, EO-WM and dynamic tasking. Novelty must come from a demonstrable observation-value method or evaluation capability.

Did the coast move—or just the waterline?

● Conditional · matched surveys needed

When can two satellite views support a signed change in a surveyed mean-sea-level beach contour?



A documented job, not imagined savings

8–10 hours of manual screening at difficult Barter Island in the USGS report; setup could take weeks. Software learning/development contributed. No savings from our method have been measured.

Data starting point

Published sandy-beach surveys. Narrabeen and San Diego Dryad metadata are CC0; imagery, tides/waves, matching and derivative rights still need clearance.

What success would have to show

More supported signed-change claims at the same false-claim rate than CoastSat/CoastSeg QC and conservative correlated physical-error bounds.

Why it might fail

Insufficient close-time survey matches; incompatible datums; or simple physical bounds are equally effective. Reef-lined Waikīkī is not a sandy-beach test.

Space scope: curated paired evidence or bounded numeric inputs → change interval / abstention → uncertainty explanations and reproducible export. No global erosion or attribution claim.

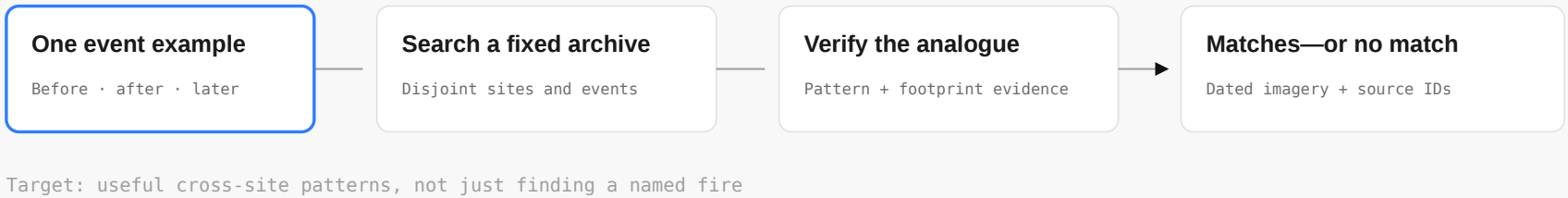
Already exists: CoastSat/CoastSeg, benchmark comparisons, tide-aware extraction and confidence displays. The proposed difference is the validity of the physical conclusion.

03 / Scientific bet

Find historical analogues—or say “no match”.

● Conditional · novelty and truth weak

Can one dated burn example find verified cross-site patterns better than cheap spectral, shape and metadata search?



The possible contribution

A benchmark for rare, verified cross-site patterns with genuinely absent galleries. Generic archive search alone is not enough.

Data starting point

MTBS and Landsat, with separately cleared assets and independently judged relevance. Start with observable burn shapes—not shared cause.

What success would have to show

Better verified top-five results than metadata, spectral-change and shape filters, without accepting absent event patterns.

Why it might fail

Cheap search already works; relevance is subjective; or there are too few null cases. A four-query pilot cannot establish archive-wide reliability.

Space scope: search a small pinned historical archive and inspect dated source footprints. Not global live search, causal diagnosis or a calibrated probability from raw similarity.

Already exists: composed EO retrieval, OpenDPR and Earth Index. A public temporal-search demo also exists; its synthetic corpus is not real-world validation.

Two important questions—but not the first build

04 Restoration recovery ● Field-data joins unresolved

Recovery—or just a wet year?

Need: independently measured water tables, exact locations/dates, weather and suitable references.

Missing: verified joins and reusable external wells; some records are on request.

Do not infer: causal restoration effects, carbon or biodiversity gains from satellite greenness.

05 Flood-area evidence audit ● Independent truth unresolved

A better map—or a better area estimate?

Need: a fixed probability-sampling budget and synchronized independent reference evidence.

Missing: expert SAR masks/model-derived labels are not automatically independent physical truth.

Do not infer: new scientific value from another flood-segmentation model alone.

Common lesson: the bottleneck is often the validity of the evidence—not model size. A polished map cannot repair missing ground truth.

Where our existing Clean-JEPA idea belongs

`Representation`

Sensor-grounded features

Test real S1/S2 availability, not channel hiding in fused annual embeddings.

`Prediction`

Future state + uncertainty

A Gaussian latent marginal does not establish semantic alignment or calibration.

`Information action`

Choose the next evidence

Value is task-specific score improvement, not generic confidence alone.

`Safe updates`

Compatibility checks

Retain Q2 as infrastructure. Reject an upgrade when task evidence is insufficient.

Earn the demo, one gate at a time

Recommended next action: metadata-only feasibility for forecast + observe. If chronology, future truth or rights fail, stop that formulation; coastal matched-survey feasibility is a separate alternative.

01 / First

Check the data contract

Source rights, native scene dates, target observability, compatible datums and preprocessing.

02 / Before outcomes

Freeze the experiment

Geographic/future-time holdouts, full input ledger, baselines, metrics and kill criteria.

03 / Then

Run a small falsifier

At most 500 declared records. No extra histories or correlated pixels disguised as independent samples.

04 / Only if earned

Release a bounded Space

Verified replay first; supported new inputs later. Show failure, abstention and provenance.

Bet	Proposed pilot ledger	What it cannot establish
Forecast + observe	480 input/verification slots; 12 sites × 1 cutoff × 4 availability replays × 10 slots	Reliable calibration from four test sites; confirmed data availability
Coastal claims	480 scored records: 320 endpoints + 160 pair claims; at most 80 scene clusters	480 independent images; blind new-coast validation
Burn search	460 scored units: 204 observation slots + 256 query/gallery judgments	Reliable no-match behavior from only two absent queries

All ledgers are proposals. Feasible dates, source approvals and accepted splits are not established. Tiny pilots can reject a bad bet; they cannot certify an operational product.

Stop or pivot if: source/chronology/independent truth fails; gains disappear against cheap baselines; success relies on future information; or uncertainty is too wide to conclude. “Inconclusive” is not a win.

Sources and how to read this visual

This is a visual translation of the existing Markdown research, not new experimental evidence. Source links are optional; all charts and controls work without a network connection.

Evidence vocabulary

Documented	Supported by a cited source or metadata check—not necessarily reproduced here.
Proposed	A hypothesis, workflow, ledger or threshold to test.
Unresolved	A gate has not passed. It is not permission to train or publish.

Important limits

No user-demand validation, saved-hours measurement, benchmark result or adoption claim. A historical replay is not live inference. Field surveys do not automatically license imagery or ancillary products. No NASA-level impact prediction.

The seven briefs behind the visual

1. Parent synthesis and recommendation
2. Forecast + observation audit
3. Coastal physical-change audit
4. Historical analogue-search audit
5. Restoration-recovery audit
6. Flood-area evidence audit
7. Sceptical cross-topic review

+ Selected primary evidence and prior art

Based on the existing synthesis, five independent audits and one sceptical review · 13 October 2026. Hypotheses and reviewer judgments—not experimental results. No dataset/model download, training or deployment.